Squares, Cubes and Roots

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS square number *(tau pūrua)* · cube number *(tau pūtoru)* · square root *(pūtakerua)*

A Square numbers ●○○

Remember: To square a number, multiply it by itself: $7^2 = 7 \times 7 = 49$.

1. Work out.

a) 4^2 = _____ **b)** 7^2 = _____ **c)** 9^2 = _____ **d)** 1^2 = _____

e) 11^2 = _____ **f)** 6^2 = _____ **g)** 12^2 = _____ **h)** 10^2 = _____

i) 8^2 = _____ **j)** 20^2 = _____ **k)** $5^2 + 3^2$ = _____ **l)** $10^2 - 6^2$ = _____

B Cube numbers ●○○

Watch out: 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

2. Work out.

a) 2^3 = _____ **b)** 3^3 = _____ **c)** 1^3 = _____ **d)** 5^3 = _____

e) 4^3 = _____ **f)** 10^3 = _____ **g)** $3^3 + 2^3$ = _____ **h)** $5^3 - 4^3$ = _____

C Square or cube? ●●○

3. Is each number a square number, a cube number, both or neither?

a) 25 _____ **b)** 27 _____ **c)** 50 _____ **d)** 64 _____

e) 8 _____ **f)** 100 _____ **g)** 125 _____ **h)** 1 _____

D Square roots ●●○

Remember: The square root undoes squaring: $\sqrt{49} = 7$ because $7^2 = 49$.

4. Work out.

a) $\sqrt{25}$ = _____ **b)** $\sqrt{81}$ = _____ **c)** $\sqrt{4}$ = _____ **d)** $\sqrt{100}$ = _____

e) $\sqrt{36}$ = _____ **f)** $\sqrt{121}$ = _____ **g)** $\sqrt{1}$ = _____ **h)** $\sqrt{64}$ = _____

i) $\sqrt{144}$ = _____ **j)** $\sqrt{400}$ = _____ **k)** $\sqrt{9} + \sqrt{16}$ = _____ **l)** $\sqrt{9 + 16}$ = _____

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E Cube roots ●●○

⚠ **Watch out:** The cube root is not "divide by 3": $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$.

5. Work out.

a) $\sqrt[3]{8}$ = _____ **b)** $\sqrt[3]{27}$ = _____ **c)** $\sqrt[3]{1}$ = _____ **d)** $\sqrt[3]{125}$ = _____

e) $\sqrt[3]{64}$ = _____ **f)** $\sqrt[3]{1000}$ = _____ **g)** $\sqrt[3]{8} + \sqrt{25}$ = ____ **h)** $\sqrt{49} - \sqrt[3]{27}$ = ____

F Estimating roots ●●●

💡 **Remember:** Find the square numbers on each side: $\sqrt{30}$ is between $\sqrt{25} = 5$ and $\sqrt{36} = 6$.

6. Say which two whole numbers each square root lies between, and which of them it is closer to.

a) $\sqrt{30}$ **b)** $\sqrt{12}$ **c)** $\sqrt{50}$

d) $\sqrt{90}$ **e)** $\sqrt{70}$ **f)** $\sqrt{5}$

7. Say which two whole numbers each cube root lies between.

a) $\sqrt[3]{20}$ **b)** $\sqrt[3]{50}$ **c)** $\sqrt[3]{100}$

8. A calculator gives these values. Round each one as shown.

a) $\sqrt{30} = 5.47722\ldots$ (1 decimal place)

b) $\sqrt{7} = 2.64575\ldots$ (2 decimal places)

c) $\sqrt{85} = 9.21954\ldots$ (1 decimal place)

★ Extension

9. A square has an area of 169 cm². How long is each side?

10. A cube has a volume of 125 cm³. How long is each edge?

11. Find two square numbers that add up to 100.

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Squares, Cubes and Roots

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS square number *(tau pūrua)* · cube number *(tau pūtoru)* · square root *(pūtakerua)*

A Square numbers ●○○

Remember: To square a number, multiply it by itself: $7^2 = 7 \times 7 = 49$.

1. Work out.

a) 5^2 = _____ **b)** 8^2 = _____ **c)** 6^2 = _____ **d)** 1^2 = _____

e) 12^2 = _____ **f)** 9^2 = _____ **g)** 11^2 = _____ **h)** 10^2 = _____

i) 7^2 = _____ **j)** 30^2 = _____ **k)** $4^2 + 6^2$ = _____ **l)** $9^2 - 5^2$ = _____

B Cube numbers ●○○

Watch out: 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

2. Work out.

a) 3^3 = _____ **b)** 2^3 = _____ **c)** 1^3 = _____ **d)** 4^3 = _____

e) 5^3 = _____ **f)** 10^3 = _____ **g)** $4^3 + 2^3$ = _____ **h)** $5^3 - 3^3$ = _____

C Square or cube? ●●○

3. Is each number a square number, a cube number, both or neither?

a) 36 _____ **b)** 8 _____ **c)** 40 _____ **d)** 1 _____

e) 27 _____ **f)** 81 _____ **g)** 64 _____ **h)** 125 _____

D Square roots ●●○

Remember: The square root undoes squaring: $\sqrt{49} = 7$ because $7^2 = 49$.

4. Work out.

a) $\sqrt{16}$ = _____ **b)** $\sqrt{49}$ = _____ **c)** $\sqrt{9}$ = _____ **d)** $\sqrt{100}$ = _____

e) $\sqrt{64}$ = _____ **f)** $\sqrt{144}$ = _____ **g)** $\sqrt{1}$ = _____ **h)** $\sqrt{81}$ = _____

i) $\sqrt{121}$ = _____ **j)** $\sqrt{900}$ = _____ **k)** $\sqrt{36} + \sqrt{64}$ = _____ **l)** $\sqrt{36 + 64}$ = _____

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E Cube roots ●●○

⚠ **Watch out:** The cube root is not "divide by 3": $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$.

5. Work out.

a) $\sqrt[3]{27}$ = _____ **b)** $\sqrt[3]{8}$ = _____ **c)** $\sqrt[3]{1}$ = _____ **d)** $\sqrt[3]{64}$ = _____

e) $\sqrt[3]{125}$ = _____ **f)** $\sqrt[3]{1000}$ = _____ **g)** $\sqrt[3]{27} + \sqrt{36}$ = ____ **h)** $\sqrt{81} - \sqrt[3]{64}$ = ____

F Estimating roots ●●●

💡 **Remember:** Find the square numbers on each side: $\sqrt{30}$ is between $\sqrt{25} = 5$ and $\sqrt{36} = 6$.

6. Say which two whole numbers each square root lies between, and which of them it is closer to.

a) $\sqrt{40}$ **b)** $\sqrt{20}$ **c)** $\sqrt{60}$

d) $\sqrt{110}$ **e)** $\sqrt{75}$ **f)** $\sqrt{8}$

7. Say which two whole numbers each cube root lies between.

a) $\sqrt[3]{30}$ **b)** $\sqrt[3]{10}$ **c)** $\sqrt[3]{90}$

8. A calculator gives these values. Round each one as shown.

a) $\sqrt{40} = 6.32455\ldots$ (1 decimal place)

b) $\sqrt{11} = 3.31662\ldots$ (2 decimal places)

c) $\sqrt{95} = 9.74679\ldots$ (1 decimal place)

★ Extension

9. A square has an area of 196 cm². How long is each side?

10. A cube has a volume of 64 cm³. How long is each edge?

11. Find two square numbers that add up to 25.

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